SHIVANG NAGESH MEDHEKAR

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SUMMARY

AI Full Stack Software Engineer with experience building production-grade AI systems, including retrieval-augmented generation (RAG), agent-based workflows, and scalable backend services. Strong background in cloud-native architectures, full-stack development, and delivering reliable AI solutions from design to deployment.

RELATED SKILLS & TECHNOLOGY

Languages	Python, JavaScript
Generative AI	OpenAI Agents SDK, Gemini, LangChain, LangGraph, LangSmith, AWS Bedrock
Front-End	React, Next.js, TypeScript, HTML, CSS
Back-End	FastAPI, Flask, Node.js, JSON, RESTful APIs, SQL, NoSQL
Devops & Cloud	AWS, GCP, GitHub Actions, Docker, Kubernetes, Terraform
Technical Skills	Agentic AI, LLM's, RAG, Distributed Systems, Event-Driven Architectures
Professional Skills	End-to-End Ownership & Delivery, Problem-Solving, Cross-Functional Collaboration

WORK EXPERIENCE

Lexim AI (lexim.ai) — February 2024 – Present
Full Stack Engineer — *Remote – Hoboken, NJ*

· Project: Praxis AI Platform

- Designed and built a scalable AI platform supporting multi-agent workflows and retrieval pipelines across multiple internal use cases.
- Implemented state-driven agent execution and RAG, improving response accuracy and reducing repeated processing.
- Integrated platform-level guardrails and validation mechanisms, increasing source-grounded responses and system reliability.
- Deployed Dockerized backend services using AWS Fargate and implemented asynchronous execution pipelines with AWS SNS, SQS, and Lambda to support hundreds of AI tasks per day.
- Deployed and operated services on AWS using ECS with Fargate and EC2 capacity providers, enabling flexible scaling and cost-efficient execution.

· Project: Agentic AI

- Designed and implemented agentic AI systems using multi-step state-driven workflows to decompose complex tasks into planning, execution, and verification phases.
- Built role-based multi-agent architectures with specialized agents for retrieval, reasoning, and validation to improve task reliability.
- Integrated retrieval-augmented generation (RAG) and semantic search to provide context-aware, source-grounded responses.
- Implemented agentic memory mechanisms to persist context across executions, reducing redundant reasoning and improving multi-step task continuity.
- Added guardrails including reflection loops, response validation, and human-in-the-loop review to reduce hallucinations and improve output accuracy.

- Integrated multiple LLM providers including Google Gemini, OpenAI, and Perplexity, with tool-calling and API orchestration to select models dynamically per task.

· Project: Lexim Query

- Built a conversational AI chatbot powered by a document-centric retrieval-augmented generation (RAG) pipeline for querying regulatory and enterprise documents.
- Implemented automated document ingestion where newly added files are embedded and indexed into AWS Knowledge Bases in near real time.
- Exposed retrieval and query orchestration through backend APIs to support semantic search and chat-based interaction.
- Integrated vector-based retrieval to ground LLM responses in source documents.
- Reduced document discovery and lookup time by ~50% for end users.

· Project: Billing Platform for AI Agents

- Designed and implemented the database schema to track agent usage, transactions, balances, and billing cycles with strong data consistency.
- Built AWS Lambda functions to calculate per-transaction costs, manage pending vs completed charges, and aggregate usage into billing cycles.
- Implemented per-agent cost attribution based on usage metrics to enable accurate and transparent AI agent billing.
- Integrated Stripe for payments, handling subscriptions and one-time charges, with secure webhook processing to reconcile payment events.
- Developed Stripe webhook handlers to process payment lifecycle events (e.g., successful charges, failures, refunds) and synchronize billing state.
- Orchestrated an event-driven billing pipeline using AWS SNS and SQS to process usage and payment events asynchronously and reliably.
- Implemented automated reconciliation, balance updates, and audit-ready billing records to support scalable, production-grade billing workflows.

· Project: Notifications for AI Agents

- Designed and implemented an event-driven notification system to deliver agent-generated alerts and system events reliably.
- Built AWS Lambda functions to process notification events and apply routing and delivery logic.
- Orchestrated an asynchronous pipeline using AWS SNS and SQS to ensure scalable, decoupled notification processing.
- Integrated Amazon SES to send transactional email notifications triggered by AI agent activity and system events.
- Designed a database-backed notification store to persist events and surface in-app notifications within the application UI.
- Ensured reliable delivery and retry handling for notifications across multiple channels.

· Project: RegIntel (Regulatory Intelligence Platform)

- Led development of the flagship regulatory intelligence platform delivering AI-driven insights across complex regulatory datasets.

- Built a React-based frontend to provide an intuitive, high-performance user experience for exploring regulatory data and insights.
- Developed backend services using Flask for core application logic and FastAPI for AI-driven and asynchronous services.
- Designed and maintained a MySQL database to support structured regulatory data, user activity, and system metadata.
- Integrated AI services to power intelligent search, analysis, and agent-based workflows within the platform.

KeelWorks Foundation (keelworks.org) May 2023 - February 2024
Frontend Engineer *Remote - Jersey City, NJ*

· Project: KeelCompass

- Led frontend development of an internal knowledge base application used to centralize and surface organizational knowledge.
- Built a React-based web application with a focus on usability, performance, and maintainable component architecture.
- Translated Figma designs into responsive, production-ready UI components.
- Collaborated with stakeholders to define information architecture and improve content discoverability.

· Project: KeelHub

- Designed and implemented a centralized hub application used to navigate and access multiple internal tools and applications.
- Contributed to UI and UX design to create a consistent, intuitive entry point across the internal product ecosystem.
- Implemented centralized authentication using Google OAuth (SSO) to provide secure, seamless access across applications.
- Improved user onboarding and access management by unifying identity and navigation under a single platform.

Inceps Consultancy Pvt. Ltd. (inceps.com) June 2020 – July 2021
Software Engineer *Onsite - Mumbai, MH, IND*

· Project: Internal ERP & Workflow Modernization

- Implemented and enhanced ERP-driven internal workflows, improving operational efficiency by 30% and reducing task completion time from 2 hours to 1.5 hours.
- Improved technical issue resolution processes, reducing average resolution time from 24 to 18 hours and increasing overall system uptime.
- Developed and deployed AWS Lambda functions to handle asynchronous processing, reducing infrastructure costs and improving response times by 30%.
- Refactored legacy frontend codebases to improve maintainability, reduce technical debt, and increase execution efficiency, resulting in a 25% performance improvement.

EDUCATION

Stevens Institute of Technology August 2021 - December 2023
MS in Computer Science *Hoboken, NJ*

University of Mumbai August 2016 - May 2020
B.E. in Computer Engineering *Mumbai, MH, IND*